
USACE / NAVFAC / AFCEC UFGS-23 05 15 (May 2022)
Change 2 - 08/24

Preparing Activity: NAVFAC Superseding
UFGS-23 05 15 (February 2014)

UNIFIED FACILITIES GUIDE SPECIFICATIONS

References are in agreement with UMRL dated April 2026

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 05 15

COMMON PIPING FOR HVAC

05/22, CHG 2: 08/24

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 GENERAL REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Material and Equipment Qualifications
 - 1.4.2 Alternative Qualifications
 - 1.4.3 Service Support
 - 1.4.4 Manufacturer's Nameplate
 - 1.4.5 Modification of References
 - 1.4.5.1 Definitions
 - 1.4.5.2 Administrative Interpretations
- 1.5 DELIVERY, STORAGE, AND HANDLING
- 1.6 ELECTRICAL REQUIREMENTS
- 1.7 ELECTRICAL INSTALLATION REQUIREMENTS
 - 1.7.1 New Work
 - 1.7.2 Modifications to Existing Systems
 - 1.7.3 High Efficiency Motors
 - 1.7.3.1 High Efficiency Single-Phase Motors
 - 1.7.3.2 High Efficiency Polyphase Motors
 - 1.7.4 Three-Phase Motor Protection
- 1.8 INSTRUCTION TO GOVERNMENT PERSONNEL
- 1.9 ACCESSIBILITY

PART 2 PRODUCTS

- 2.1 ELECTRICAL HEAT TRACING
- 2.2 PIPE AND FITTINGS
 - 2.2.1 Type BCS, Black Carbon Steel
 - 2.2.2 Type BCS-125, 862 kilopascal Service 125-psi Service
 - 2.2.3 Type GCS, Galvanized Carbon Steel
 - 2.2.4 Type GCS-DWV, Galvanized Steel Drain, Waste and Vent
 - 2.2.5 Type CISP-DWV, Cast-Iron Drain, Waste and Vent

- 2.2.6 Type CPR, Copper
 - 2.2.6.1 Type CPR-A, Copper Above Ground
 - 2.2.6.2 Type CPR-U, Copper Under Ground
 - 2.2.6.3 Type CPR-INS, Copper Under Ground Insulated
- 2.2.7 Polypropylene Pipe
- 2.2.8 Grooved Pipe Couplings and Fittings
- 2.3 PIPING SPECIALTIES
 - 2.3.1 Air Separator
 - 2.3.2 Air Vents
 - 2.3.3 Compression Tank
 - 2.3.4 Dielectric Connections
 - 2.3.5 Expansion Vibration Isolation Joints
 - 2.3.6 Flexible Pipe
 - 2.3.7 Flexible Metallic Pipe
 - 2.3.8 Flexible Metal Steam Hose
 - 2.3.9 Metallic Expansion Joints
 - 2.3.10 Hose Faucets
 - 2.3.11 Pressure Gages
 - 2.3.12 Sight-Flow Indicators
 - 2.3.13 Sleeve Couplings
 - 2.3.14 Thermometers
 - 2.3.15 Pump Suction Strainers
 - 2.3.16 Line Strainers, Water Service
 - 2.3.17 Line Strainers, Steam Service
- 2.4 VALVES
 - 2.4.1 Ball and Butterfly Valves
 - 2.4.2 Drain, Vent, and Gage Cocks
 - 2.4.3 Gate Valves (GAV)
 - 2.4.4 Globe and Angle Valves (GLV-ANV)
 - 2.4.5 Standard Check Valves (SCV)
 - 2.4.6 Nonslam Check Valves (NSV)
- 2.5 MISCELLANEOUS MATERIALS
 - 2.5.1 Bituminous Coating
 - 2.5.2 Bolting
 - 2.5.3 Elastomer Caulk
 - 2.5.4 Escutcheons
 - 2.5.5 Flashing
 - 2.5.6 Flange Gaskets
 - 2.5.7 Grout
 - 2.5.8 Pipe Thread Compounds
- 2.6 SUPPORTING ELEMENTS
 - 2.6.1 Building Structure Attachments
 - 2.6.1.1 Anchor Devices, Concrete and Masonry
 - 2.6.1.2 Beam Clamps
 - 2.6.1.3 Type 19 and 23 C-Clamps
 - 2.6.1.4 Inserts, Concrete
 - 2.6.2 Horizontal Pipe Attachments
 - 2.6.2.1 Single Pipes
 - 2.6.2.2 Parallel Pipes
 - 2.6.3 Vertical Pipe Attachments
 - 2.6.4 Hanger Rods and Fixtures
 - 2.6.5 Supplementary Steel

PART 3 EXECUTION

- 3.1 PIPE INSTALLATION
- 3.2 VALVES
- 3.3 SUPPORTING ELEMENTS INSTALLATION
- 3.4 PENETRATIONS

- 3.5 SLEEVES
- 3.6 ESCUTCHEONS
- 3.7 FLASHINGS
- 3.8 UNDERGROUND PIPING INSTALLATION
- 3.9 HEAT TRACE CABLE INSTALLATION
- 3.10 DISINFECTION
- 3.11 HEAT TRACE CABLE TESTS
- 3.12 OPERATION AND MAINTENANCE
- 3.13 PAINTING OF NEW EQUIPMENT
 - 3.13.1 Factory Painting Systems
 - 3.13.2 Shop Painting Systems for Metal Surfaces

-- End of Section Table of Contents --

USACE / NAVFAC / AFCEC

UFGS-23 05 15 (May 2022)

Change 2 - 08/24

Preparing Activity: NAVFAC

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SECTION 23 05 15

COMMON PIPING FOR HVAC
05/22, CHG 2: 08/24

NOTE: This guide specification covers the requirements for standard basic mechanical work and should be supplemented by use of other mechanical sections as required.

Show on the drawings detailed upstream and downstream piping anchor provisions.

Install flexible metallic pipe vertically to keep dirt out of convolutions.

Coordinate design detail and specification for each installation with the manufacturer to ensure that length, stiffness of hose, and slack are suitable for the intended offset, travel, and imposed service under normal and shock conditions.

Indicate on the drawings use of flexible metal steam for main steamline dripping where amount of expansion and contraction is such that movement cannot be readily accommodated by piping configuration, with excessive stress on pressurized components or where there is a tendency to cause leaks at connections to mains. Tunnels, trenches, manholes, and above-ground steamlines are prime locations; ensure the pressure rating accounts for water-hammer shock. This specification is limited to **15 millimeter through 25 millimeter 1/2 inch through 1 inch**. Use welded pipe, valve, and hole connections wherever possible. Provide a welded end steam strainer upstream of hose to prevent welding bead penetration of bellows upon start up. Wherever possible, install flexible metal steam hose vertically.

Show on the Drawings, or supplement the specifications to include, calculated movement of piping, operating pressure and temperature ranges, fluid velocity, piping anchor and guiding

provisions, limit stops, installation length, end connections, and special conditions such as angular displacement and vibration analysis in one or more planes.

This specification does not include slip-type expansion joints or ball joints.

Adhere to [UFC 1-300-02 Unified Facilities Guide Specifications \(UFGS\) Format Standard](#) when editing this guide specification or preparing new project specification sections. Edit this guide specification for project specific requirements by adding, deleting, or revising text. For bracketed items, choose applicable item(s) or insert appropriate information.

Remove information and requirements not required in respective project, whether or not brackets are present.

Comments, suggestions and recommended changes for this guide specification are welcome and should be submitted as a [Criteria Change Request \(CCR\)](#).

PART 1 GENERAL

1.1 REFERENCES

NOTE: This paragraph is used to list the publications cited in the text of the guide specification. The publications are referred to in the text by basic designation only and listed in this paragraph by organization, designation, date, and title.

Use the Reference Wizard's Check Reference feature when you add a Reference Identifier (RID) outside of the Section's Reference Article to automatically place the reference in the Reference Article. Also use the Reference Wizard's Check Reference feature to update the issue dates.

References not used in the text will automatically be deleted from this section of the project specification when you choose to reconcile references in the publish print process.

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325

(2017) Steel Construction Manual

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.18.1/CSA B125.1	(2024) Plumbing Supply Fittings
ASME A112.19.2/CSA B45.1	(2024) Standard for Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals
ASME B1.20.7	(1991; R 2013) Standard for Hose Coupling Screw Threads (Inch)
ASME B1.21M	(1997; R 2013; R 2018; R 2023) Standard for Metric Screw Threads - MJ Profile
ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.4	(2021) Gray Iron Threaded Fittings; Classes 125 and 250
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2021) Forged Fittings, Socket-Welding and Threaded
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.25	(2022) Buttwelding Ends
ASME B16.26	(2024) Cast Copper Alloy Fittings for Flared Copper Tubes
ASME B16.34	(2021) Valves - Flanged, Threaded and Welding End
ASME B16.39	(2025) Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300
ASME B31.3	(2024) Process Piping
ASME B36.10M	(2022) Welded and Seamless Wrought Steel Pipe
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications
ASME BPVC SEC VIII D1	(2023) BPVC Section VIII-Rules for

Construction of Pressure Vessels Division 1

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M (2019) Specification for Filler Metals for
Brazing and Braze Welding

AWS WHB-2.9 (2004) Welding Handbook; Volume 2, Welding
Processes, Part 1

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M (2026) Standard Specification for General
Requirements for Rolled Structural Steel
Bars, Plates, Shapes, and Sheet Piling

ASTM A53/A53M (2024) Standard Specification for Pipe,
Steel, Black and Hot-Dipped, Zinc-Coated,
Welded and Seamless

ASTM A74 (2025) Standard Specification for Cast
Iron Soil Pipe and Fittings

ASTM A105/A105M (2026) Standard Specification for Carbon
Steel Forgings for Piping Applications

ASTM A106/A106M (2026) Standard Specification for Seamless
Carbon Steel Pipe for High-Temperature
Service

ASTM A126 (2004; R 2023) Standard Specification for
Gray Iron Castings for Valves, Flanges,
and Pipe Fittings

ASTM A183 (2014; R 2020) Standard Specification for
Carbon Steel Track Bolts and Nuts

ASTM A197/A197M (2020; R 2023; E 2023) Standard
Specification for Cupola Malleable Iron

ASTM A216/A216M (2021) Standard Specification for Steel
Castings, Carbon, Suitable for Fusion
Welding, for High-Temperature Service

ASTM A234/A234M (2025) Standard Specification for Piping
Fittings of Wrought Carbon Steel and Alloy
Steel for Moderate and High Temperature
Service

ASTM A276/A276M (2025) Standard Specification for
Stainless Steel Bars and Shapes

ASTM A278/A278M (2001; R 2024) Standard Specification for
Gray Iron Castings for Pressure-Containing
Parts for Temperatures Up to 650 degrees F
(350 degrees C)

ASTM A307 (2023) Standard Specification for Carbon
Steel Bolts, Studs, and Threaded Rod 60

000 PSI Tensile Strength

ASTM A312/A312M	(2022a) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A480/A480M	(2025b) Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM A563	(2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts
ASTM A563M	(2007; R 2013) Standard Specification for Carbon and Alloy Steel Nuts (Metric)
ASTM A888	(2024) Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B62	(2017; R 2025) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B88M	(2020) Standard Specification for Seamless Copper Water Tube (Metric)
ASTM B117	(2025) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B370	(2022) Standard Specification for Copper Sheet and Strip for Building Construction
ASTM B749	(2025) Standard Specification for Lead and Lead Alloy Strip, Sheet and Plate Products
ASTM C67/C67M	(2025) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
ASTM C109/C109M	(2024) Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or (50-mm) Cube Specimens)
ASTM C404	(2024) Standard Specification for Aggregates for Masonry Grout
ASTM C476	(2023) Standard Specification for Grout for Masonry
ASTM C553	(2024) Standard Specification for Mineral Fiber Blanket Thermal Insulation for

Commercial and Industrial Applications

ASTM C564	(2026) Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM D2000	(2018; R 2024) Standard Classification System for Rubber Products in Automotive Applications
ASTM D2308	(2007; R 2013) Standard Specification for Thermoplastic Polyethylene Jacket for Electrical Wire and Cable
ASTM E1	(2014; R 2025) Standard Specification for ASTM Liquid-in-Glass Thermometers
ASTM E84	(2026) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E814	(2024) Standard Test Method for Fire Tests of Penetration Firestop Systems
ASTM F104	(2011; R 2020) Standard Classification System for Nonmetallic Gasket Materials
ASTM F568M	(2007) Standard Specification for Carbon and Alloy Steel Externally Threaded Metric Fasteners
ASTM F2389	(2024a) Standard Specification for Pressure-rated Polypropylene (PP) Piping Systems

CAST IRON SOIL PIPE INSTITUTE (CISPI)

CISPI 310	(2012) Coupling for Use in Connection with Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
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FLUID SEALING ASSOCIATION (FSA)

FSA-0017	(1995e6) Standard for Non-Metallic Expansion Joints and Flexible Pipe Connectors Technical Handbook
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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 515	(2017) Standard for the Testing, Design, Installation, and Maintenance of Electrical Resistance Heat Tracing for Industrial Applications
IEEE C2	(2023; ERTA 1 2023; ERTA 2-3 2025)

National Electrical Safety Code

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

MSS SP-58	(2025) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
MSS SP-67	(2022) Butterfly Valves
MSS SP-70	(2011) Gray Iron Gate Valves, Flanged and Threaded Ends
MSS SP-72	(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service
MSS SP-80	(2019) Bronze Gate, Globe, Angle, and Check Valves
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends
MSS SP-125	(2010) Gray Iron and Ductile Iron In-Line, Spring-Loaded, Center-Guided Check Valves

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 00001	(2024) Motors and Generators
NEMA MG 10	(2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors
NEMA MG 11	(1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code
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NSF INTERNATIONAL (NSF)

NSF/ANSI 14	(2024) Plastics Piping System Components and Related Materials
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U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-C-18480	(1982; Rev B; Notice 2 2009; Notice 3 2022) Coating Compound, Bituminous, Solvent, Coal-Tar Base
MIL-DTL-17813	(2009; Rev H; Supp 1 2009; Notice 1 2013; Notice 2 2018; Notice 3 2023) Expansion

Joints, Pipe, Metallic Bellows, General
Specification for

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-1922	(Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Caulking Anchors, Single Lead)
CID A-A-1923	(Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Lag, Machine and Externally Threaded Wedge Bolt Anchors)
CID A-A-1924	(Rev A; Notice 3) Shield, Expansion (Self Drilling Tubular Expansion Shell Bolt Anchors)
CID A-A-1925	(Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield Expansion (Nail Anchors)
CID A-A-55614	(Basic; Notice 2) Shield, Expansion (Non-Drilling Expansion Anchors)
CID A-A-55615	(Basic; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Wood Screw and Lag Bolt Self-Threading Anchors)

UL SOLUTIONS (UL)

UL 1479	(2015; Reprint Mar 2026) UL Standard for Safety Fire Tests of Penetration Firestops
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1.2 GENERAL REQUIREMENTS

NOTE: If Section 23 30 00 HVAC AIR DISTRIBUTION is not included in the project specification, applicable requirements thereof should be inserted and the first paragraph deleted. If Section 23 05 48.19 SEISMIC BRACING FOR HVAC is not included in the project specification, applicable requirements thereof should be inserted and the second paragraph deleted. If Section 40 05 13.96 WELDING PROCESS PIPING is not included in the project specification, applicable requirements thereof should be inserted and the third paragraph deleted.

- [a. Section 23 30 00 HVAC AIR DISTRIBUTION applies to work specified in this section
-] [b. Section 23 05 48.19 SEISMIC BRACING FOR HVAC applies to work specified in this section.
-] [c. Section 40 05 13.96 WELDING PROCESS PIPING applies to work specified in this section.
-] d. Submit [Records of Existing Conditions](#) consisting of the results of

Contractor's survey of work area conditions and features of existing structures and facilities within and adjacent to the jobsite. Commencement of work constitutes acceptance of the existing conditions.

- e. Include with [Equipment Foundation Data](#) for piping systems all plan dimensions of foundations and relative elevations, equipment weight and operating loads, horizontal and vertical loads, horizontal and vertical clearances for installation, and size and location of anchor bolts.
- f. Submit [Fabrication Drawings](#) for pipes, valves and specialties consisting of fabrication and assembly details to be performed in the factory.
- g. Submit [Material, Equipment, and Fixture Lists](#) for pipes, valves and specialties including manufacturer's style or catalog numbers, specification and drawing reference numbers, warranty information, and fabrication site information. Provide a complete list of construction equipment to be used.
- h. Submit [Manufacturer's Standard Color Charts](#) for pipes, valves and specialties showing the manufacturer's recommended color and finish selections.
- i. Include with [Listing of Product Installations](#) for piping systems identification of at least 5 units, similar to those proposed for use, that have been in successful service for a minimum period of 5 years. Include in the list purchaser, address of installation, service organization, and date of installation.
- j. Submit [Record Drawings](#) for pipes, valves and accessories providing current factual information including deviations and amendments to the drawings, and concealed and visible changes in the work.
- k. Submit [Connection Diagrams](#) for pipes, valves and specialties indicating the relations and connections of devices and apparatus by showing the general physical layout of all controls, the interconnection of one system (or portion of system) with another, and internal tubing, wiring, and other devices.
- l. Submit [Coordination Drawings](#) for pipes, valves and specialties showing coordination of work between different trades and with the structural and architectural elements of work. Detail all drawings sufficiently to show overall dimensions of related items, clearances, and relative locations of work in allotted spaces. Indicate on drawings where conflicts or clearance problems exist between various trades.

1.3 SUBMITTALS

NOTE: Review Submittal Description (SD) definitions in Section 01 33 00 SUBMITTAL PROCEDURES and edit the following list, and corresponding submittal items in the text, to reflect only the submittals required for the project. The Guide Specification technical editors have classified those items that require Government approval, due to their complexity or criticality, with a "G." Generally, other submittal items can be reviewed by the Contractor's

Quality Control System. Only add a "G" to an item, if the submittal is sufficiently important or complex in context of the project.

For Army projects, fill in the empty brackets following the "G" classification, with a code of up to three characters to indicate the approving authority. Codes for Army projects using the Resident Management System (RMS) are: "AE" for Architect-Engineer; "DO" for District Office (Engineering Division or other organization in the District Office); "AO" for Area Office; "RO" for Resident Office; and "PO" for Project Office. Codes following the "G" typically are not used for Navy and Air Force projects.

The "S" classification indicates submittals required as proof of compliance for sustainability Guiding Principles Validation or Third Party Certification and as described in Section 01 33 00 SUBMITTAL PROCEDURES.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Material, Equipment, and Fixture Lists[; G, [____]]

SD-02 Shop Drawings

Record Drawings[; G, [____]]

Connection Diagrams[; G, [____]]

Coordination Drawings[; G, [____]]

Fabrication Drawings[; G, [____]]

Installation Drawings[; G, [____]]

SD-03 Product Data

Pipe and Fittings[; G, [____]]

Piping Specialties[; G, [____]]

Valves[; G, [____]]

Miscellaneous Materials[; G, [____]]

Supporting Elements[; G, [____]]

Equipment Foundation Data[; G, [____]]

SD-04 Samples

Manufacturer's Standard Color Charts[; G, [____]]

SD-05 Design Data

Pipe and Fittings[; G, [____]]

Piping Specialties[; G, [____]]

Valves[; G, [____]]

SD-06 Test Reports

Hydrostatic Tests[; G, [____]]

Air Tests[; G, [____]]

Valve-Operating Tests[; G, [____]]

Drainage Tests[; G, [____]]

Pneumatic Tests[; G, [____]]

Non-Destructive Electric Tests[; G, [____]]

System Operation Tests[; G, [____]]

SD-07 Certificates

Record of Satisfactory Field Operation[; G, [____]]

List of Qualified Permanent Service Organizations[; G, [____]]

Listing of Product Installations[; G, [____]]

Records of Existing Conditions[; G, [____]]

Surface Resistance[; G, [____]]

Shear and Tensile Strengths[; G, [____]]

Temperature Ratings[; G, [____]]

Bending Tests[; G, [____]]

Flattening Tests[; G, [____]]

Transverse Guided Weld Bend Tests[; G, [____]]

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals[; G, [____]]

1.4 QUALITY ASSURANCE

1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Provide standard products in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use includes applications of equipment and materials under similar circumstances and of similar size. Ensure the product has been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.4.2 Alternative Qualifications

Products having less than a two-year field service record are acceptable if a certified [record of satisfactory field operation](#) for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.4.3 Service Support

Ensure the equipment items are supported by service organizations. Submit a certified [list of qualified permanent service organizations](#) for support of the equipment which includes their addresses and qualifications. Select service organizations that are reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.4.4 Manufacturer's Nameplate

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable.

1.4.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer.

1.4.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions are considered mandatory, the word "should" is interpreted as "shall." Reference to the "code official" is interpreted to mean the "Contracting Officer." For Navy owned property, interpret references to the "owner" to mean the "Contracting Officer." For leased facilities, references to the "owner" is interpreted to mean the "lessor." References to the "permit holder" are interpreted to mean the "Contractor."

1.4.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of

Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, are applied as appropriate by the Contracting Officer and as authorized by his administrative cognizance and the FAR.

1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

[1.6 ELECTRICAL REQUIREMENTS

**NOTE: Use this paragraph for other than
SOUTHNAVFACENGCOM projects.**

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Ensure motors, controllers, disconnects and contactors conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors is not permitted. Provide controllers and contactors with a maximum of 120 volt control circuits, and auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, include the cost of additional electrical service and related work under the section that specified that motor or equipment. Provide power wiring and conduit for field installed equipment under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

]1.7 ELECTRICAL INSTALLATION REQUIREMENTS

**NOTE: Use this paragraph and its subparagraphs
regarding electrical components and energy efficient
motors for SOUTHNAVFACENGCOM projects.**

Ensure electrical installations conform to IEEE C2, NFPA 70, and requirements specified herein.

1.7.1 New Work

Provide electrical components of mechanical equipment, such as motors, motor starters [(except starters/controllers which are indicated as part of a motor control center)], control or push-button stations, float or pressure switches, solenoid valves, integral disconnects, and other devices functioning to control mechanical equipment, as well as control wiring and conduit for circuits rated 100 volts or less, to conform with the requirements of the section covering the mechanical equipment. Extended voltage range motors are not permitted. Provide under Division 26, the interconnecting power wiring and conduit, control wiring rated 120 volts (nominal) and conduit, [the motor control equipment forming a part

of motor control centers,] and the electrical power circuits, except internal wiring for components of package equipment is provided as an integral part of the equipment. When motors and equipment furnished are larger than sizes indicated, provide any required changes to the electrical service as may be necessary and related work as a part of the work for the section specifying that motor or equipment.

1.7.2 Modifications to Existing Systems

Where existing mechanical systems and motor-operated equipment require modifications, provide electrical components under Division 26.

1.7.3 High Efficiency Motors

1.7.3.1 High Efficiency Single-Phase Motors

Unless otherwise specified, provide high efficiency single-phase fractional-horsepower alternating-current motors corresponding to the applications listed in NEMA MG 11.

1.7.3.2 High Efficiency Polyphase Motors

Unless otherwise specified, select polyphase motors based on high efficiency characteristics relative to the applications as listed in NEMA MG 10. Additionally, ensure polyphase squirrel-cage medium induction motors with continuous ratings meet or exceed energy efficient ratings in accordance with Table 12-6C of NEMA MG 00001.

1.7.4 Three-Phase Motor Protection

Provide controllers for motors rated one 1.34 kilowatts one horsepower and larger with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

]1.8 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Provide instructors thoroughly familiar with all parts of the installation and trained in operating theory as well as practical operation and maintenance work.

Give instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished is as specified in the individual section. When more than four man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.9 ACCESSIBILITY

NOTE: The following requirement is intended to solicit the installer's help in the prudent location of equipment when he has some control over locations. However, designers should not rely on it at all since enforcing this requirement in the field would be difficult. Therefore, the system designer needs to layout and indicate the locations of equipment, control devices, and access doors so that most of the accessibility questions are resolved inexpensively during design.

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

2.1 ELECTRICAL HEAT TRACING

Provide heat trace systems for pipes, valves, and fittings that are in accordance with **IEEE 515** and be UL listed. System include all necessary components, including heaters and controls to prevent freezing.

Provide self-regulating heaters consisting of two 16 AWG tinned-copper bus wires embedded in parallel in a self-regulating polymer core that varies its power output to respond to temperature along its length. Ensure heater is able to be crossed over itself without overheating. Obtain approval before used directly on plastic pipe. Cover heater with a radiation cross-linked modified polyolefin dielectric jacket in accordance with **ASTM D2308**.

[For installation on plastic piping, apply the heater using aluminum tape. Provide heater with an outer braid of tinned-copper and an outer jacket of modified polyolefin in accordance with **ASTM D2308**, to provide a good ground path and to enhance the heater's ruggedness.

] *****

NOTE: Self-regulation factor is defined as the percentage reduction, without thermostatic control, of the heater output going from **4 degrees C 40 degrees F** pipe temperature operation to **66 degrees C 150 degrees F** pipe temperature operation.

Provide heater with self-regulating factor of at least [90] [_____] percent, in order to provide energy conservation and to prevent overheating.

Operate heater on line voltages of [120] [208] [220] [240] [277] volts without the use of transformers.

NOTE: Required heater output rating is in watts per meter at **10 degrees C** foot at **50 degrees F**. Heater

selection based on 25 millimeter one-inch fiberglass insulation on metal piping.

Size Heater according to the following table:

Pipe Size (DN)

(Millimeter Diameter)	Minus 23 degrees C	Minus 29 degrees C
80 or less	16 watts per meter (wpm)	16 watts per meter (wpm)
100	16 wpm	26 wpm
150	26 wpm	26 wpm
200	2 strips/16 wpm	2 strips/26 wpm
300 to 356	2 strips/26 wpm	2 strips/26 wpm

Pipe Size

(Inch, Diameter)	Minus 10 degrees F	Minus 20 degrees F
3 inches or less	5 watts per foot (wpf)	5 wpf
4 inch	5 wpf	8 wpf
6 inch	8 wpf	8 wpf
8 inch	2 strips/5 wpf	2 strips/8 wpf
12 inch	2 strips/8 wpf	2 strips/8 wpf

Control systems by an ambient sensing thermostat set at 4 degrees C 40 degrees F either directly or through an appropriate contactor.

2.2 PIPE AND FITTINGS

NOTE: Do not use press fittings on NAVFAC projects. Do not use grooved connections on NAVFAC projects.

Submit equipment and performance data for pipe and fittings consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.2.1 Type BCS, Black Carbon Steel

NOTE: This pipe is applicable for chilled, hot, dual-temperature, and cooling-tower water. Refer to Specification Section 23 64 26 for system details.

Ensure pipe DN6 through DN300 1/8 through 12 inches is Schedule 40 black carbon steel, conforming to ASTM A53/A53M.

Ensure pipe DN6 through DN250 1/8 through 10 inches is Schedule 40 seamless or electric-resistance welded black carbon steel, conforming to ASTM A53/A53M, [Type E, Grade B (electric-resistance welded)] [Type S (seamless)]. Grade A should be used for permissible field bending, in both cases.

Ensure pipe DN300 through DN610 12 through 24 inches is 9.52 millimeter 0.375-inch wall seamless black carbon steel, conforming to ASTM A53/A53M, [Type E, Grade B (electric-resistance welded)] [Type S (seamless)].

Ensure fittings DN50 and under 2 inches and under are 1034 kilopascal 150-pounds per square inch, gage (psig) working steam pressure (wsp) banded black malleable iron screwed, conforming to ASTM A197/A197M and ASME B16.3.

Ensure unions DN50 and under 2 inches and under are 1724 kilopascal 250 pounds per square inch, wsp female, screwed, black malleable iron with brass-to-iron seat, and ground joint, conforming to ASME B16.39.

Ensure fittings DN65 and over 2-1/2 inches and over are Steel butt weld, conforming to ASTM A234/A234M and ASME B16.9 to match pipe wall thickness.

Ensure flanges DN65 and over 2-1/2 inches and over are 1034 kilopascal 150-pound forged-steel conforming to ASME B16.5, welding neck to match pipe wall thickness.

2.2.2 Type BCS-125, 862 kilopascal Service 125-psi Service

NOTE: This pipe is applicable for steam- and condensate-piping systems at pressures less than 862 kilopascal 125 pounds per square inch (psi). Avoid screwed-end connections in condensate piping wherever possible. See Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS for black carbon steel pipe for higher pressure ratings.

Ensure pipe DN6 through DN40 1/8 through 1-1/2 inches is Schedule 40 steam, Schedule 80 condensate, furnace butt weld, black carbon steel, conforming to ASTM A53/A53M, Type F (furnace butt welded, continuous welded) and ASME B36.10M.

Ensure pipe DN50 through DN250 2 through 10 inches is Schedule 40 steam, Schedule 80 condensate, seamless or electric-resistance welded black carbon steel, conforming to ASTM A53/A53M [Type E, Grade B (electric-resistance welded)] [Type S (seamless)] and ASME B36.10M.

**NOTE: For condensate piping, modify following (for
DN300 12 inches and over) to schedule 40 or schedule
80, if necessary.**

- Ensure pipe DN300 through DN610 12 through 24 inches is 9.52 millimeter 0.375-inch wall, [seamless] [electric-resistance] welded black carbon steel, conforming to ASTM A53/A53M [Type E, Grade B (electric-resistance welded)] [Type S (seamless) and ASME B36.10M].
- [Ensure fittings DN50 and under 2 inches and under are 862 kilopascal 125-psig wsp, cast iron, screwed end, conforming to ASTM A126 Class A and ASME B16.4.
-][Ensure fittings DN50 and under 2 inches and under are 1034 kilopascal 150-psig wsp banded black malleable iron screwed, conforming to ASTM A197/A197M and ASME B16.3.
-][Ensure fittings DN25 through DN50 1 through 2 inches are 14 or 21 megapascal 2,000-or 3,000-psi water, oil, or gas (wog) to match pipe wall, forged carbon steel socket weld, conforming to ASTM A105/A105M and ASME B16.11.
-][Ensure fittings DN50 and under 2 inches and under are 862 kilopascal 125-psig wsp, cast iron, screwed end, conforming to ASTM A126 Class A and ASME B16.4.
-][Ensure fittings DN65 and over 2-1/2 inches and over are wall thickness to match pipe, long radius butt weld, black carbon steel, conforming to ASTM A234/A234M, Grade WPB and ASME B16.9.
-][Ensure couplings DN50 and under 2 inches and under are commercial standard weight for Schedule 40 pipe and commercial extra heavy weight for Schedule 80 pipe, black carbon steel where threaded, and 14 or 21 megapascal 2,000-or 3,000-psi wog forged carbon steel, conforming to ASTM A105/A105M and ASME B16.11, where welded.
-][Ensure flanges DN65 and over 2-1/2 inches and over are 1035 kilopascal, 150-pound, forged carbon-steel welding neck, with raised face or flat face and concentric serrated finish, conforming to ASTM A105/A105M and ASME B16.5.
-][Conform grooved pipe couplings and fittings in accordance with paragraph GROOVED PIPE COUPLINGS AND FITTINGS.
-]2.2.3 Type GCS, Galvanized Carbon Steel

**NOTE: This pipe is applicable for potable water and
rain water leader systems.**

Ensure pipe DN15 through DN250, and where indicated 1/2 through 10 inches, and where indicated is Schedule 40 seamless or electric-resistance welded galvanized steel conforming to ASTM A53/A53M, Type E, Grade B (electric-resistance welded) or Type S (seamless).

Ensure pipe DN300 and over 12 inches and over is 9.52 millimeter 0.375-inch

wall, seamless, galvanized steel, conforming to ASTM A53/A53M, Grade B.

Ensure fittings DN50 and under 2 inches and under are 1034 kilopascal 150-psig wsp banded galvanized malleable iron screwed, conforming to ASTM A197/A197M and ASME B16.3.

Ensure unions DN50 and under 2 inches and under are 1034 kilopascal 150-psig wsp female, screwed, galvanized malleable iron with brass-to-iron seat and ground joint.

Ensure fittings DN65 and over 2-1/2 inches and over are 862 kilopascal 125-psig wsp cast-iron flanges and flanged fittings, conforming to ASTM A126, Class A and ASME B16.1.

Conform grooved pipe couplings and fittings in accordance with paragraph GROOVED PIPE COUPLINGS AND FITTINGS.

As an option, use 1034 kilopascal 150-psig wsp banded galvanized malleable iron screwed fittings, conforming to ASTM A197/A197M and ASME B16.3.

2.2.4 Type GCS-DWV, Galvanized Steel Drain, Waste and Vent

NOTE: Nonferrous piping exposed to view in finished spaces and normally chrome plated is specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE. Select A53 pipe where bending and flattening tests are required.

Ensure pipe (all sizes) is Schedule 40 [seamless] [electric-resistance welded] galvanized carbon steel, conforming to ASTM A53/A53M, Grade A.

Furnace butt weld pipe is acceptable for sizes less than DN50 2 inches.

[Provide risers DN80 3 inches and larger are Type CISP-DWV.

][Ensure fittings are galvanized, [coated] [uncoated], screwed, cast iron, recessed pattern drainage fittings, conforming to ASTM A126.

][Use long radius fittings wherever space permits. Short-turn tees, branches, and ells may be used for vent piping and connections of branch lines to battery fixtures, except wall-hung water closets.

]2.2.5 Type CISP-DWV, Cast-Iron Drain, Waste and Vent

NOTE: When project requires risers DN80 3 inches and larger, include Type CISP-DWV materials specification.

[Provide soil pipe drain, waste, and vent bell-and-spigot type pipe cast iron, conforming to ASTM A74.[Joints shall be two-gasket system type neoprene, conforming to ASTM C564.][Caulk and lead all joints in lines to provide proper leaktight support and alignment.] Select the extra heavy (CISP-DWV-XH) pipe class.

][Provide soil pipe drain, waste, and vent hubless cast iron pipe and

fittings, conforming to **ASTM A888**. Joints shall be heavyweight no-hub couplings with stainless steel clamps, conforming to **CISPI 310**.

]2.2.6 Type CPR, Copper

NOTE: Copper pipe above ground and below ground is acceptable for chilled, hot, dual-temperature, cooling-tower water, and potable-water systems. Refer to Sections 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS and 22 00 00 PLUMBING, GENERAL PURPOSE for system details.

2.2.6.1 Type CPR-A, Copper Above Ground

Ensure tubing **DN50 and under 2 inches and under** is seamless copper tubing, conforming to **ASTM B88M, ASTM B88**, Type L (hard-drawn for all horizontal and all exposed vertical lines, annealed for concealed vertical lines).

Ensure fittings **DN50 and under 2 inches and under** are **1034 kilopascal 150-psig** wsp wrought-copper solder joint fittings conforming to **ASME B16.22**.

Ensure unions **DN50 and under 2 inches and under** are **1034 kilopascal 150-psig** wsp wrought-copper solder joint, conforming to **ASME B16.22**.

[Provide brazing rod with Classification BCuP-5, conforming to **AWS A5.8/A5.8M**.

][Use solder, alloy Sb-5, conforming to **ASTM B32**.

]2.2.6.2 Type CPR-U, Copper Under Ground

NOTE: For sizes under DN80 3 inches.

Provide Type K seamless copper tube piping, conforming to **ASTM B88M ASTM B88**. Use wrought copper socket-joint fittings, conforming to **ASME B16.22**. Ensure fittings for connection to corporation cocks are cast bronze, flared-type, conforming to **ASME B16.26**. Braze the joints.

2.2.6.3 Type CPR-INS, Copper Under Ground Insulated

NOTE: Type CPR-INS material is commercially available in sizes to and including DN105 4 inches OD.

Since pipe is protected from soil by insulation system, Type L copper tube may be used if suitable for water carried at a cost saving of 10 percent.

Type CPR-INS material may be used for hot water supply and return connected to tunnel mains.

Provide insulated Type K seamless copper tube piping conforming to **ASTM B88M ASTM B88**. Use wrought copper socket-joint fittings, conforming

to ASME B16.22. Braze the joints.

Provide insulation not less than DN50 2 inches thick, suitable for continuous service temperatures of not less than 121 degrees C 250 degrees F. Use factory-molded, closed-cell polyurethane foam insulation of not less than 40 kilogram per cubic meter 2.5 pounds per cubic foot density. Waterproof insulation with an extruded rigid Type II virgin polyvinylchloride, with minimum wall thickness of 1.52 millimeter through 102 millimeter 60 mils through 4 inches outside diameter, 2.16 millimeter through 168.28 millimeter 85 mils through 6.625 inches and 2.79 millimeter through 273 millimeter 110 mils through 12.750 inches. Provide fitting covers fabricated from the same materials and thickness as adjacent pipe covering according to the manufacturer's directions.

2.2.7 Polypropylene Pipe

Pipe is manufactured from a PP-R resin meeting the short-term properties and long-term strength requirements of ASTM F2389. Pipe is made in a three layer extrusion process. Piping contains a fiber layer (faser) to restrict thermal expansion. Pipe complies with the rated pressure requirements of ASTM F2389. Ensure layers are incorporated in the pipe wall to limit thermal expansion to 3.38 cm per degree C per meter 2 1/4-inches per 100 F per 100-ft. If the hydronic system includes ferrous components, an oxygen barrier is required in pipe wall.

Ensure pipe is certified by NSF International as complying with NSF/ANSI 14, and ASTM F2389.

Ensure pipe wrap or insulation meets the requirements of ASTM E84. Ensure the system has a Flame Spread Classification of less than 25 and Smoke Development rating of less than 50.

Where pipe is exposed to direct UV light for more than 30 days, provide a Factory applied, UV-resistant coating or alternative UV protection.

2.2.8 Grooved Pipe Couplings and Fittings

NOTE: Do not use grooved connections on NAVFAC projects.

Provide housing for all couplings, fabricated in two or more parts, of black, ungalvanized malleable iron castings. Ensure coupling gasket is molded synthetic rubber, conforming to ASTM D2000. Ensure coupling bolts are oval-neck, track-head type, with hexagonal heavy nuts conforming to ASTM A183.

Fabricate all pipe fittings used with couplings of black, ungalvanized malleable iron castings. Where a manufacturer's standard-size malleable iron fitting pattern is not available, approved fabricated fittings may be used.

Fabricate fittings from Schedule 40 or 19 millimeter 0.75-inch wall ASTM A53/A53M, Grade B seamless steel pipe; long radius seamless welding fittings with wall thickness to match pipe, conforming to ASTM A234/A234M and ASME B16.9.

2.3 PIPING SPECIALTIES

Submit equipment and performance data for **piping specialties** consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.3.1 Air Separator

Air separated from converter discharge water is ejected by a reduced-velocity device vented to the compression tank.

[Provide a commercially constructed separator, designed and certified to separate not less than 80 percent of entrained air on the first passage of water and not less than 80 percent of residual on each successive pass. Provide shop drawings detailing all piping connections proposed for this work.

[Ensure the air separator is carbon steel, designed, fabricated, tested, and stamped in conformance with **ASME BPVC SEC VIII D1** for service pressures not less than **862 kilopascal 125 psi**.

2.3.2 Air Vents

[Provide manual air vents using **10 millimeter 3/8-inch** globe valves.

[*****
NOTE: This size vent is suitable for most systems, and passes 9.40 liter per second of free air 20 cubic feet of free air per minute at a system pressure of 862 kilopascal 125 psi. Where a system is filled at a certain rate, larger vents or a multiple assembly with safety features should be used.

[Provide automatic air vents on pumps, mains, and where indicated using ball-float construction. Ensure the vent inlet is not less than **DN20 3/4-inch ips** and the outlet not less than **8 millimeter 1/4-inch ips**. Orifice size is **3 millimeter 1/8 inch**. Provide corrosion-resistant steel trim conforming to [**ASTM A276/A276M**] [**ASTM A480/A480M**]. Fit vent with try-cock. Ensure vent discharges air at any pressure up to **1034 kilopascal 150 psi**. Ensure outlet is copper tube routed.

2.3.3 Compression Tank

Provide compression tank designed, fabricated, tested, and stamped for a working pressure of not less than **862 kilopascal 125 psi** in accordance with **ASME BPVC SEC VIII D1**. Ensure tank is hot-dip galvanized after fabrication to produce not less than **51 grams 1.5 ounces** of zinc coating per square **meter foot** of single-side surface.

Tank accessories include red-lined gage-glass complete with glass protectors and shutoff valves, air charger and drainer, and manual vent.

2.3.4 Dielectric Connections

Electrically isolate dissimilar pipe metals from each other by couplings, unions, or flanges commercially manufactured for that purpose and rated for the service pressure and temperature.

2.3.5 Expansion Vibration Isolation Joints

NOTE: Drawings should show detailed piping anchor provisions where expansion vibration isolation joints are used.

This joint may also serve as a dielectric connector.

Construct single or multiple arch-flanged expansion vibration isolation joints of steel-ring reinforced chloroprene-impregnated cloth materials. Design joint to absorb the movement of the pipe sections in which installed with no detrimental effect on the pipe or connected equipment. Back flanges with ferrous-metal backing rings. Provide control rod assemblies to restrict joint movement. Coat all nonmetallic exterior surfaces of the joint with chlorosulphonated polyethylene. Provide grommets in limit bolt hole to absorb noise transmitted through the bolts.

NOTE: If other elastomers are substituted for chloroprene, temperature limits may be lowered to 82 degrees C 180 degrees F or less.

Ensure joints are suitable for continuous-duty working temperature of at least 121 degrees C 250 degrees F.

NOTE: Select the following paragraph where solids accumulating in arch would cause cutting of carcass. Note that all movements will be reduced by 50 percent.

Fill arches with soft chloroprene.

Ensure joint, single-arch, movement limitations and size-related, pressure characteristics conform to FSA-0017.

2.3.6 Flexible Pipe

NOTE: Drawings should show detailed upstream and downstream piping anchor provisions and location with respect to axis of motion where flexible pipe is used.

Grooved couplings and vibration-isolated pipe hangers should be considered.

Flexible pipe may also serve as a dielectric connector.

Select following paragraph for manufacturer's standard-service pipe.

Construct flexible pipe vibration and pipe-noise eliminators of wire-reinforced, rubber-impregnated cloth and cord materials and be flanged. Back the flanges with ferrous-metal backing rings. Ensure service pressure-rating is a minimum 1.5 times actual service, with surge pressure at 82 degrees C 180 degrees F.

NOTE: Anticipated life of chloroprene units at 121 degrees C 250 degrees F is 5 to 10 years.

Construct flexible pipe vibration and pipe noise eliminators of wire-reinforced chloroprene-impregnated cloth and cord materials. Ensure the pipe is flanged. Provide all flanges backed with ferrous-metal backing rings. Coat nonmetallic exterior surfaces of the flexible pipe with an acid- and oxidation-resistant chlorosulphonated polyethylene. Rate the flexible pipe for continuous duty at 896 kilopascal and 121 degrees C 130 psi and 250 degrees F.

Ensure unit pipe lengths, face-to-face, are not less than the following:

NOTE: The following lengths are basic recommendations: each application should be reviewed for optimum length.

<u>INSIDE DIAMETER (DN)</u>	<u>UNIT PIPE LENGTH</u>
[To 65, inclusive	305 millimeter
80 to 100, inclusive	450 millimeter
125 to 300, inclusive	600 millimeter]
[To 80, inclusive	450 millimeter
110 to 250, inclusive	600 millimeter
300 and larger	914 millimeter]
<u>INSIDE DIAMETER</u>	<u>UNIT PIPE LENGTH</u>
[To 2-1/2 inches, inclusive	12 inches
3 to 4 inches, inclusive	18 inches
5 to 12 inches, inclusive	24 inches]
[To 3 inches, inclusive	18 inches
4 to 10 inches, inclusive	24 inches
12 inches and larger	36 inches]

INSIDE DIAMETER

UNIT PIPE LENGTH

2.3.7 Flexible Metallic Pipe

Ensure flexible pipe is the bellows-type with wire braid cover and designed, constructed, and rated in accordance with the applicable requirements of [ASME B31.3](#).

Minimum working pressure rating is [\[345\] \[690\] kilopascal at 149 degrees C](#)
[\[50\] \[100\] psi at 300 degrees F](#).

[Ensure minimum burst pressure is [four] [_____] times working pressure at [149 degrees C](#) [300 degrees F](#). Bellows material is AISI Type 316L corrosion-resistant steel. Ensure braid is AISI 300 series corrosion-resistant steel wire.

][Ensure welded end connections are Schedule 80 carbon steel pipe, conforming to [ASTM A106/A106M](#), Grade [B] [C].

][Provide threaded end connections; hex-collared Schedule 40, AISI Type 316L corrosion-resistant steel, conforming to [ASTM A312/A312M](#).

][Ensure flanged end connection rating and materials conform to specifications for system primary-pressure rating.

2.3.8 Flexible Metal Steam Hose

Provide a bellows type hose with wire braid cover and designed, constructed, and rated in accordance with the applicable requirements of [ASME B31.3](#).

Ensure the working steam pressure rating is [862 kilopascal at 260 degrees C](#)
[125 psi at 500 degrees F](#).

[Ensure minimum burst pressure is [four] [_____] times working steam pressure at [149 degrees C](#) [300 degrees F](#).

] Ensure bellows material is AISI Type 316L corrosion-resistant steel. Braid is AISI Type 300-series corrosion-resistant steel wire.

[Provide welded end connections; Schedule 80 carbon steel pressure tube, conforming to [ASTM A106/A106M](#), Grade [B] [C].

][Provide threaded end connections; hex-collared Schedule 40, AISI Type 316L corrosion-resistant steel, conforming to [ASTM A312/A312M](#).

][Ensure flanged end connection rating and materials conform to specifications for system primary-pressure rating.

2.3.9 Metallic Expansion Joints

[Provide metallic-bellows expansion joints conforming to [MIL-DTL-17813](#).

][Provide Type I expansion joints; (corrugated bellows, unreinforced), [Class 1 (single bellows, expansion joint)], [Class 2 (double bellows, expansion joint)].

] Design and construct joints to absorb all of the movements of the pipe

sections in which installed, with no detrimental effect on pipe or supporting structure.

Rate, design, and construct joints for pressures to 862 kilopascal 125 psig and temperatures to 260 degrees C 500 degrees F.

Ensure joints have a designed bursting strength in excess of [four] [_____] times their rated pressure.

Ensure joints are capable of withstanding a hydrostatic test of 1.5 times their rated pressure while held at their uncompressed length without leakage or distortion that may adversely affect their life cycle.

Ensure life expectancy is not less than 10,000 cycles.

Ensure movement capability of each joint exceeds calculated movement of piping by [100] [_____] percent.

Provide bellows and internal sleeve material of AISI Type 304, 304L, or 321 corrosion-resistant steel.

End connections require no field preparation other than cleaning.

[Butt weld end preparation of expansion joints conform to the same codes and standards requirements as applicable to the piping system materials at the indicated joint location.

][Flanges of flanged-end expansion joints conforms to the same codes and standard requirements as are applicable to companion flanges specified for the given piping system at the indicated joint location.

] Provide joints, DN65 2-1/2 inches and smaller, with internal guides and limit stops.

Provide joints, DN80 3 inches and larger, with removable external covers, internal sleeves, and purging connection. Size sleeves to accommodate lateral clearance required, with minimum reduction of flow area, and with oversized bellows where necessary. When a sleeve requires a gasket as part of a locking arrangement, provide the gasket used by the manufacturer. Joints without purging connection may be provided; however, remove these from the line prior to, or not installed until, cleaning operations are complete.

[Provide the cylindrical end portion of the reinforced bellows element with a thrust sleeve of sufficient thickness to bring that portion within applicable code-allowable stress. Provide 360 degrees support for the element and end-reinforcing ring with the sleeve.

][Ensure expansion joints have four, equidistant, permanent tram points clearly marked on each joint end. Locate points to prevent obliteration during installation. Include distance between tram points indicating installed lengths in shop drawings. Overall dimension after joint installation is subject to approval from the Contracting Officer.

] Ensure each expansion joint has adjustable clamps or yokes provided at quarter points, straddling the bellows. Overall joint length is set by the manufacturer to maintain joints in manufacturer's recommended position during installation.

NOTE: Securely anchor pipe lines containing expansion joints to completely resist the thrust due to the pressure acting on the full internal area of the corrugations. Also, properly guide the pipe to prevent misalignment of the joint. Correlate details of anchors and guides for each application.

Permanently and legibly mark each joint with the manufacturer's name or trademark and serial number; the size, series, or catalog number; bellows material; and directional-flow arrow.

2.3.10 Hose Faucets

NOTE: Normally delete vacuum breaker when faucets are installed in non-potable-water lines.

Construct hose faucets with 15 millimeter 1/2 inch male inlet threads, hexagon shoulder, and 20 millimeter 3/4 inch hose connection, conforming to ASME A112.18.1/CSA B125.1. Ensure hose-coupling screw threads conform to ASME B1.21M ASME B1.20.7.

Provide vandal proof, atmospheric-type vacuum breaker on the discharge of all potable water lines.

2.3.11 Pressure Gages

Ensure pressure gages conform to ASME B40.100 and to requirements specified herein. Pressure-gage size is 90 millimeter 3-1/2 inches nominal diameter. Ensure case is corrosion-resistant steel, conforming to any of the AISI 300 series of ASTM A6/A6M, with an ASM No. 4 standard commercial polish or better. Equip gages with adjustable red marking pointer and damper-screw adjustment in inlet connection. Align service-pressure reading at midpoint of gage range. Ensure all gages are Grade B or better and be equipped with gage isolators.

NOTE: Retain the following paragraph only if pressure gages are used on steam piping.

[Fit steam gages with black steel syphons and steam service pressure-rated gage cocks or valves.

]2.3.12 Sight-Flow Indicators

Construct sight-flow indicators for pressure service on 80 millimeter 3-inch ips and smaller of bronze with specially treated single- or double-glass sight windows and have a bronze, nylon, or tetrafluoroethylene rotating flow indicator mounted on an AISI Type [304] [316] corrosion-resistant steel shaft. Body may have screwed or flanged end. Provide pressure- and temperature-rated assembly for the applied service. Flapper flow-type indicators are not acceptable.

2.3.13 Sleeve Couplings

Sleeve couplings for plain-end pipe consist of one steel middle ring, two steel followers, two chloroprene or Buna-N elastomer gaskets, and the necessary steel bolts and nuts.

2.3.14 Thermometers

Ensure thermometers conform to **ASTM E1**, except for being filled with a red organic liquid. Provide an industrial pattern armored glass thermometer, (well-threaded and seal-welded). Ensure thermometers installed **1800 millimeter 6 feet** or higher above the floor have an adjustable angle body. Ensure scale is not less than **180 millimeter 7 inches** long and the case face is manufactured from manufacturer's standard polished aluminum or AISI 300 series polished corrosion-resistant steel. Thermometer range is [_____]. Provide thermometers with nonferrous separable wells. Provide lagging extension to accommodate insulation thickness.

2.3.15 Pump Suction Strainers

NOTE: To preclude cavitation, check the following conditions prior to specifying: NPSH, flow rate, open area, screen size, and pressure drop across strainer.

Provide a cast iron strainer body, rated for not less than **172 kilopascal at 38 degrees C 25 psig at 100 degrees F**, with flanges conforming to **ASME B16.1**, Class 125. Strainer construction is such that there is a machined surface joint between body and basket that is normal to the centerline of the basket.

Ensure minimum ratio of open area of each basket to pipe area is 3 to 1. Provide a basket with AISI 300 series corrosion-resistant steel wire mesh with perforated backing.

Ensure mesh is capable of retaining all particles larger than 1,000 micrometer, with a pressure drop across the strainer body of not more than **5 kilopascal 0.5 psi** when the basket is two-thirds dirty at maximum system flow rate. Provide reducing fittings from strainer-flange size to pipe size.

Provide a [differential-pressure gage][pressure gage with **2 kilopascal 0.25-pound** graduations] fitted with a two-way brass cock across the strainer.

Provide manual air vent cocks in cap of each strainer.

2.3.16 Line Strainers, Water Service

Provide Y-type strainers with removable basket. Ensure strainers in sizes **DN50 2-inch ips** and smaller have screwed ends; in sizes **DN65 2-1/2-inch ips** and larger, strainers have flanged ends. Ensure body working-pressure rating exceeds maximum service pressure of installed system by at least 50 percent. Ensure body has cast-in arrows to indicate direction of flow. Ensure all strainer bodies fitted with screwed screen retainers have straight threads and gasketed with nonferrous metal. For strainer bodies **DN65 2-1/2-inches** and larger, fitted with bolted-on screen retainers,

provide offset blowdown holes. Fit all strainers larger than DN65 2-1/2-inches with manufacturer's standard ball-type blowdown valve. Ensure body material is [cast bronze conforming to ASTM B62][cast iron conforming to Class 30 ASTM A278/A278M]. Where system material is nonferrous, use nonferrous metal for the metal strainer body material.

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 1.14 millimeter 0.045-inch. Ensure strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is[AISI Type [304] [316] corrosion-resistant steel][Monel metal].

2.3.17 Line Strainers, Steam Service

Provide Type Y strainers with removable strainer element.

Use flanged body end connections for all valves larger than DN50 2 inches, unless butt weld ends are specified. Use [screwed] [socket] weld for sizes DN50 2 inches and under to suit specified piping system end connection and maintenance requirements[or be welded].

For strainers located in tunnels, trenches, manholes, and valve pits, use welded end connections.

Body working steam pressure rating is the same as the primary valve rating for system in which strainer is installed, except where welded end materials requirements result in higher pressure ratings. Ensure body has integral cast or forged arrows to indicate direction of flow. Provide strainer bodies with blowdown valves that have discharge end plugged with a solid metal plug. Make closure assembly with tetrafluoroethylene tape. Ensure bodies fitted with bolted-on screen retainers have offset blowdown holes.

Body materials are [cast steel conforming to ASTM A216/A216M, Grade WCB] [forged carbon steel conforming to ASTM A105/A105M] [manufacturer's standard metallurgical equivalents for service pressures of 1035 kilopascal 150-psi wsp and greater, and for lower pressure ratings where welding is required] [cast iron conforming to ASTM A126, Class B, for service pressures 862 kilopascal 125-psi wsp and less].

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 0.51 millimeter 0.020 inch or equivalent wire mesh. Strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is AISI Type [304] [316] corrosion-resistant steel and fitted with backup screens where necessary to prevent collapse.

2.4 VALVES

NOTE: Figure 1A is a one piece body.

Figure 1B is a vertically split body with the split to one side of the ball.

Figure 1C is a top entry.

Figure 1D is a three piece body.

Submit equipment and performance data for valves consisting of corrosion resistance and life expectancy. Submit design analysis and calculations consisting of rates of flow, head losses, inlet and outlet design, and pressure calculations. Also include in data, pipe dimensions, as well as temperature ratings, vibration and thrust limitations, minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

Polypropylene valves will comply with the performance requirements of ASTM F2389. Valves shall conform to ASME B16.34.

2.4.1 Ball and Butterfly Valves

Ensure ball valves conform to MSS SP-72 for flanged valves and MSS SP-110 for screwed-end valves for Figure [1A] 1 piece body, [1B] vertically split body, [1C] top entry [1D] three piece body and are rated for service at not less than 1207 kilopascal at 93 degrees C 175 psig at 200 degrees F. For valve bodies in sizes DN50 2 inches and smaller, use screwed-end connection-type constructed of Class A copper alloy. For valve bodies in sizes [DN50] [DN65] 2-1/2 inches and larger, use flanged-end connection type, constructed of Class [D] [E] [F] material. Balls and stems of valves DN50 2 inches and smaller are manufacturer's standard with hard chrome plating finish. Balls and stems of valves DN65 2-1/2 inches and larger are manufacturer's standard Class C corrosion-resistant steel alloy with hard chrome plating. Balls of valves DN150 6 inches and larger may be Class D with 900 Brinell hard chrome plating. Ensure valves are suitable for flow from either direction and seal equally tight in either direction. Valves with ball seals held in place by spring washers are not acceptable. Ensure all valves have adjustable packing glands. Seats and seals are fabricated from tetrafluoroethylene.

Ensure butterfly valves conform to MSS SP-67 and are the wafer type for mounting between specified flanges. Ensure valves are rated for 1034 kilopascal 150-psig shutoff and nonshock working pressure. Select bodies of cast ferrous metal conforming to ASTM A126, Class B, and to ASME B16.1 for body wall thickness. Seats and seals are fabricated from resilient elastomer designed for field removal and replacement.

2.4.2 Drain, Vent, and Gage Cocks

Provide [T-head] [lever handle] drain, vent, and gage cocks, ground key type, with washer and screw, constructed of polished ASTM B62 bronze, and rated 862 kilopascal 125-psi wsp. Ensure end connections are rated for specified service pressure.

Ensure pump vent cocks, and where spray control is required, are UL umbrella-hood type, constructed of manufacturer's standard polished brass. Ensure cocks are 15 millimeter 1/2-inch ips male, end threaded, and rated at not less than 862 kilopascal at 107 degrees C 125 psi at 225 degrees F.

2.4.3 Gate Valves (GAV)

Ensure gate valves DN50 2 inches and smaller conform to MSS SP-80. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated use union-ring bonnet, screwed-end type. Make packing of non-asbestos type materials. Use rising stem type valves.

Ensure gate valves DN65 2-1/2 inches and larger, are Type I, (solid wedge disc, tapered seats, steam rated); Class 125 (862 kilopascal 125-psig steam-working pressure at 178 degrees C 353 degrees F saturation); and 1379 kilopascal 200-psig, wog (nonshock), conforming to MSS SP-70 and to requirements specified herein. Select flanged valves, with bronze trim and outside screw and yoke (OS&Y) construction. Make packing of non-asbestos type materials.

2.4.4 Globe and Angle Valves (GLV-ANV)

Ensure globe and angle valves DN50 2 inches and smaller, are 862 kilopascal 125-pound, 125-psi conforming to MSS SP-80 and to requirements specified herein. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated, use union-ring bonnet, screwed-end type. Ensure disc is free to swivel on the stem in all valve sizes. Composition seating-surface disc construction may be substituted for all metal-disc construction. Make packing of non-asbestos type materials. Ensure disk and packing are suitable for pipe service installed.

Ensure globe and angle valves, DN65 2-1/2 inches and larger, are cast iron with bronze trim. Ensure valve bodies are cast iron conforming to ASTM A126, Class A, as specified for Class 1 valves under MSS SP-80. Select flanged valves in conformance with ASME B16.1. Valve construction is outside screw and yoke (OS&Y) type. Make packing of non-asbestos type materials.

2.4.5 Standard Check Valves (SCV)

Ensure standard check valves in sizes DN50 2 inches and smaller are 862 kilopascal 125-psi swing check valves except as otherwise specified. Provide lift checks where indicated. Ensure swing-check pins are nonferrous and suitably hard for the service. Select composition type discs. Ensure the swing-check angle of closure is manufacturer's standard unless a specific angle is needed.

Use cast iron, bronze trim, swing type check valves in sizes DN65 2-1/2 inches and larger. Ensure valve bodies are cast iron, conforming to ASTM A126, Class A and valve ends are flanged in conformance with ASME B16.1. Swing-check pin is AISI Type or approved equal corrosion-resistant steel. Angle of closure is manufacturer's standard unless a specific angle is needed. Ensure valves have bolted and gasketed covers.

Provide check valves with [external spring-loaded] [lever-weighted], positive-closure devices and valve ends are [mechanical joint] [push-on] [flanged].

2.4.6 Nonslam Check Valves (NSV)

NOTE: The following specification is adequate for most construction situations. Where unusual hydraulic conditions occur, review closing time and in-service adjustment capability of helical-coil valve construction versus other construction.

Provide check valves at pump discharges in sizes DN50 2 inches and larger with nonslam or silent-check operation conforming to MSS SP-125. Select a

valve disc or plate that closes before line flow can reverse to eliminate slam and water-hammer due to check-valve closure. Ensure valve is Class 125 rated for 1379 kilopascal 200-psi maximum, nonshock pressure at 66 degrees C 150 degrees F in sizes to DN300 12 inches. Use valves that are [wafer type to fit between flanges conforming to ASME B16.1][fitted with flanges conforming to ASME B16.1]. Valve body may be cast iron, or equivalent strength ductile iron. Select disks using manufacturer's standard bronze, aluminum bronze, or corrosion-resistant steel. Ensure pins, springs, and miscellaneous trim are manufacturer's standard corrosion-resistant steel. Disk and shaft seals are Buna-N elastomer tetrafluoroethylene.

2.5 MISCELLANEOUS MATERIALS

Submit equipment and performance data for miscellaneous materials consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

2.5.1 Bituminous Coating

Ensure the bituminous coating is a solvent cutback, heavy-bodied material to produce not less than a 0.30 millimeter 12-mil dry-film thickness in one coat, and is recommended by the manufacturer to be compatible with factory-applied coating and rubber joints.

For previously coal-tar coated and uncoated ferrous surfaces underground, use bituminous coating solvent cutback coal-tar type, conforming to MIL-C-18480.

2.5.2 Bolting

Ensure flange and general purpose bolting is hex-head and conforms to ASTM F568M, Class 4.8 or above ASTM A307, Grade B (bolts, for flanged joints in piping systems where one or both flanges are cast iron). Heavy hex-nuts conform to ASTM A563M ASTM A563. Square-head bolts and nuts are not acceptable. Ensure threads are coarse-thread series.

2.5.3 Elastomer Caulk

Use two-component polysulfide- or polyurethane-base elastomer caulking material, conforming to ASTM C920.

2.5.4 Escutcheons

Manufacture escutcheons from nonferrous metals and chrome-plated except when AISI 300 series corrosion-resistant steel is provided. Ensure metals and finish conforms to ASME A112.19.2/CSA B45.1.

Use one-piece escutcheons where mounted on chrome-plated pipe or tubing, and one-piece of split-pattern type elsewhere. Ensure all escutcheons have provisions consisting of [internal spring-tension devices] [setscrews] for maintaining a fixed position against a surface.

2.5.5 Flashing

Ensure sheetlead conforms to ASTM B749, [UNS Alloy Number L50049 (intended for use in laboratories and shops in general application)][UNS Alloy Number L51121 (for use where lead sheet of high purity and improved structural strength is indicated)].

Ensure sheet copper conforms to ASTM B370 and be not less than 4.88 kilogram per square meter 16 ounces per square foot weight.

2.5.6 Flange Gaskets

Provide compressed non-asbestos sheets, conforming to ASTM F104, coated on both sides with graphite or similar lubricant, with nitrile composition, binder rated to 399 degrees C 750 degrees F.

2.5.7 Grout

NOTE: When moisture or uncured concrete occurs, metallic grout may cause buildup of pressure that, under confinement, could be sufficient to misaligned equipment.

Provide shrink-resistant grout as a premixed and packaged metallic-aggregate, mortar-grouting compound conforming to ASTM C404 and ASTM C476.

NOTE: Specify epoxy grout, particularly where mild chemical resistance is necessary or where oil soaking may occur.

For service with acids, polyester grouts should be specified.

Where high anchor-bolt torques (2712 newton-meter) (2,000 ft-lb) are applied, epoxy polyamides will cold-flow.

Ensure shrink-resistant grout is a combination of pre-measured and packaged epoxy polyamide or amine resins and selected aggregate mortar grouting compound conforming to the following requirements:

Tensile strength		13.100 Megapascal, minimum
Compressive strength	ASTM C109/C109M	96.527 Megapascal, minimum
Shrinkage, linear		0.003 mm per millimeter, maximum
Water absorption	ASTM C67/C67M	0.1 percent, maximum
Bond strength to		6.895 Megapascal, minimum steel in shear minimum
Tensile strength		1,900 psi, minimum
Compressive strength	ASTM C109/C109M	14,000 psi, minimum
Shrinkage, linear		0.00012 inch per inch, maximum
Water absorption	ASTM C67/C67M	0.1 percent, maximum

Bond strength to

1,000 psi, minimum steel in shear
minimum

2.5.8 Pipe Thread Compounds

Use polytetrafluoroethylene tape not less than 0.05 to 0.08 millimeter 2 to 3 mils thick in potable and process water and in chemical systems for pipe sizes to and including DN25 1-inch ips. Use polytetrafluoroethylene dispersions and other suitable compounds for all other applications upon approval by the Contracting Officer; however, do not use lead-containing compounds in potable water systems.

2.6 SUPPORTING ELEMENTS

Submit equipment and performance data for the supporting elements consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

Provide all necessary piping systems and equipment supporting elements, including but not limited to: building structure attachments; supplementary steel; hanger rods, stanchions, and fixtures; vertical pipe attachments; horizontal pipe attachments; anchors; guides; and spring-cushion, variable, or constant supports. Ensure supporting elements are suitable for stresses imposed by systems pressures and temperatures and natural and other external forces normal to this facility without damage to supporting element system or to work being supported.

Ensure supporting elements conform to requirements of ASME B31.3, and MSS SP-58, except as noted.

Ensure attachments welded to pipe are made of materials identical to that of pipe or materials accepted as permissible raw materials by referenced code or standard specification.

Ensure supporting elements exposed to weather are hot-dip galvanized or stainless steel. Select materials of such a nature that their apparent and latent-strength characteristics are not reduced due to galvanizing process. Electroplate supporting elements in contact with copper tubing with copper.

Type designations specified herein are based on MSS SP-58. Ensure masonry anchor group-, type-, and style-combination designations are in accordance with CID A-A-1922, CID A-A-1923, CID A-A-1924, CID A-A-1925, CID A-A-55614, and CID A-A-55615. Provide support elements, except for supplementary steel, that are cataloged, load rated, commercially manufactured products.

2.6.1 Building Structure Attachments

**NOTE: Review specific instructions relative to
anchor devices in support elements installation
paragraph prior to selection of following text.**

2.6.1.1 Anchor Devices, Concrete and Masonry

Ensure anchor devices conform to CID A-A-1922, CID A-A-1923, CID A-A-1924, CID A-A-1925, CID A-A-55614, and CID A-A-55615

For cast-in, floor mounted, equipment anchor devices, provide adjustable positions.

[Provide built-in masonry anchor devices.

] Do not use powder-actuated anchoring devices to support any mechanical systems components.

2.6.1.2 Beam Clamps

Ensure beam clamps are center-loading **MSS SP-58** Type [20] [21] [28] [29] [30] [_____].

[When it is not possible to use center-loading beam clamps, eccentric-loading beam clamps, **MSS SP-58** Type [20] [25] [27] may be used for piping sizes **DN50 2 inches** and less and for piping sizes **DN50 through DN250 2 through 10 inches** provided two counterbalancing clamps are used per point of pipe support. Where more than one rod is used per point of pipe support, determine rod diameter in accordance with referenced standards.

]2.6.1.3 Type 19 and 23 C-Clamps

Torque Type 19 and 23 C-clamps in accordance with **MSS SP-58** and have both locknuts and retaining devices furnished by the manufacturer. Field fabricated C-clamp bodies or retaining devices are not acceptable.

2.6.1.4 Inserts, Concrete

Use concrete **MSS SP-58** Type [18] [_____] inserts. When applied to piping in sizes **DN50 2 inches ips** and larger and where otherwise required by imposed loads, insert and wire a **305 millimeter 1-foot** length of **13 millimeter 1/2-inch** reinforcing rod through wing slots. Submit proprietary-type continuous inserts for approval.

2.6.2 Horizontal Pipe Attachments

2.6.2.1 Single Pipes

Support piping in sizes to and including **DN50 2-inch ips** by **MSS SP-58** Type 6 solid malleable iron pipe rings, except that, use split-band-type rings in sizes up to **DN25 1-inch ips**.

Support piping in sizes through **DN200 8-inch ips** inclusive by **MSS SP-58** Type [1] [3] [4] attachments.

Use **MSS SP-58** Type 1 and Type 6 assemblies on vapor-sealed insulated piping and have an inside diameter larger than pipe being supported to provide adequate clearance during pipe movement.

Where thermal movement of a point in a piping system **DN100 4 inches** and larger would cause a hanger rod to deflect more than 4 degrees from the vertical or where a horizontal point movement exceeds **13 millimeter 1/2 inch**, use **MSS SP-58** Type [41] [44 through 46] [49] pipe rolls.

Support piping in sizes larger than **DN200 8-inch ips** with **MSS SP-58** Type [41] [44 through 46] [49] pipe rolls.

Use **MSS SP-58** Type 40 shields on all insulated piping. Ensure area of the

supporting surface is such that compression deformation of insulated surfaces does not occur. Roll away longitudinal and transverse shield edges from the insulation.

Provide insulated piping without vapor barrier on roll supports with [MSS SP-58](#) Type 39 saddles.

Provide spring supports as indicated.

2.6.2.2 Parallel Pipes

Use trapeze hangers fabricated from structural steel shapes, with U-bolts, in congested areas and where multiple pipe runs occur. Ensure structural steel shapes[conform to supplementary steel requirements][be of commercially available, proprietary design, rolled steel].

2.6.3 Vertical Pipe Attachments

Ensure vertical pipe attachments are [MSS SP-58](#) Type 8.

Include complete fabrication and attachment details of any spring supports in shop drawings.

2.6.4 Hanger Rods and Fixtures

Use only circular cross section rod hangers to connect building structure attachments to pipe support devices. Use pipe, straps, or bars of equivalent strength for hangers only where approved by the Contracting Officer.

Provide turnbuckles, swing eyes, and clevises as required by support system to accommodate temperature change, pipe accessibility, and adjustment for load and pitch. Rod couplings are not acceptable.

2.6.5 Supplementary Steel

Where it is necessary to frame structural members between existing members or where structural members are used in lieu of commercially rated supports, design and fabricate such supplementary steel in accordance with [AISC 325](#).

PART 3 EXECUTION

3.1 PIPE INSTALLATION

Submit certificates for pipes, valves and specialties showing conformance with test requirements as contained in the reference standards contained in this section. Provide certificates verifying [Surface Resistance](#), [Shear and Tensile Strengths](#), [Temperature Ratings](#), [Bending Tests](#), [Flattening Tests](#) and [Transverse Guided Weld Bend Tests](#).

Provide test reports for [Hydrostatic Tests](#), [Air Tests](#), [Valve-Operating Tests](#), [Drainage Tests](#), [Pneumatic Tests](#), [Non-Destructive Electric Tests](#) and [System Operation Tests](#), in compliance with referenced standards contained within this section.

Fabricate and install piping systems in accordance with [ASME B31.3](#), [MSS SP-58](#), and [AWS WHB-2.9](#).

Submit [Installation Drawings](#) for pipes, valves and specialties. Drawings include the manufacturer's design and construction calculations, forces required to obtain rated axial, lateral, or angular movements, installation criteria, anchor and guide requirements for equipment, and equipment room layout and design. Ensure drawings specifically advise on procedures to be followed and provisions required to protect expansion joints during specified hydrostatic testing operations.

NOTE: Select one of the following two paragraphs.
The first paragraph is the standard option that provides true electrical isolation. This is the preferred option. The second paragraph is a non-standard option intended to be used as a cost-saving option. This option also covers dielectric nipples that are often factory-installed on mechanical equipment. This option provides a level of electrical insulation between the two metals, but may still lead to galvanic corrosion over time as metals are not truly isolated.

- [Ensure connections between steel piping and copper piping are electrically isolated from each other. Dielectric pipe unions shall be installed to prevent galvanic corrosion. The dielectric unions shall have metal connections on both ends. The ends shall be threaded, flanged, or brazed to match adjacent piping. The metal parts of the union shall be separated so that the electrical current is below one percent of the galvanic current which would exist upon metal-to-metal contact. Gaskets, flanges, and unions shall be installed in accordance with manufacturer's recommendations.
 -][Provide dielectric nipples between steel piping and copper piping to reduce galvanic corrosion. Dielectric nipples shall be Schedule 40 galvanized steel conforming to [ASTM A53/A53M](#) with inert, non-corrosive thermoplastic lining. For pipe sizes [DN50 2 inches ips](#) smaller, a bronze ball valve can be used between dissimilar metals.
 -] Make final connections to equipment with [unions] [flanges] installed every [30480 millimeter 100 feet](#) of straight run. Install unions in the line downstream of screwed- and welded-end valves.
- Ream all pipe ends before joint connections are made.
- Make screwed joints with specified joint compound with not more than three threads showing after joint is made up.
- Apply joint compounds to the male thread only and exercise care to prevent compound from reaching the unthreaded interior of the pipe.
- Install screwed unions, welded unions, or bolted flanges wherever required to permit convenient removal of equipment, valves, and piping accessories from the piping system for maintenance.
- Securely support piping systems with due allowance for thrust forces, thermal expansion and contraction. Do not subject the system to mechanical, chemical, vibrational or other damage as specified in [ASME B31.3](#).

Ensure field welded joints conform to the requirements of the AWS WHB-2.9, ASME B31.3, and ASME BPVC SEC IX.

- [Make piping systems butt weld joints with backing rings. Use compatible backing ring materials with materials being joined. Ensure joint configuration conforms to ASME B16.25.

For polypropylene pipe, make fusion-weld joints in accordance with the pipe and fitting manufacturer's specifications and product standards. Use fusion-weld tooling, welding machines, and electrofusion devices specified by the pipe and fittings manufacturer. Prior to joining, prepare the pipe and fittings in accordance with ASTM F2389 and the manufacturer's specifications. Ensure joint preparation, setting and alignment, fusion process, cooling times and working pressure are in accordance with the pipe and fitting manufacturer's specifications.

-] *****
NOTE: Prior to selection of one of the following two paragraphs, review requirements of ASME B31.3 and ASME BPVC SEC IX to avoid conflict and redundancy. Also review PFI ES-19 and PFI ES-28 if materials specifications have been rewritten or supplemented.

- [Accomplish preheat and postheat treatment of welds in accordance with ASME BPVC SEC IX and ASME B31.3.

-][Take all necessary precautions during installation of flexible pipe and hose including flushing and purging with water, steam, and compressed air to preclude bellows failure due to pipe line debris lodged in bellows. Ensure installation conforms to manufacturer's instructions.

]3.2 VALVES

Install valves in piping mains and all branches and at equipment where indicated and as specified.

Install valves to permit isolation of branch piping and each equipment item from the balance of the system.

Install riser and downcomer drains above piping shutoff valves in piping DN65 2-1/2 inches and larger. Tap and fit shutoff valve body with a DN15 1/2-inch plugged globe valve.

Install valves unavoidably located in furred or other normally inaccessible places with access panels adequately sized for the location and located so that concealed items may be serviced, maintained, or replaced.

3.3 SUPPORTING ELEMENTS INSTALLATION

Install supporting elements in accordance with the referenced codes and standards.

Support piping from building structure. Do not support piping from roof deck or from other pipe.

Run piping parallel with the lines of the building. Space and install

piping and components so that a threaded pipe fitting may be removed between adjacent pipes and so that there is no less than **DN15 1/2 inch** of clear space between the finished surface and other work and between the finished surface of parallel adjacent piping. Arrange hangers on different adjacent service lines running parallel with each other in line with each other and parallel to the lines of the building.

Install piping support elements at intervals specified hereinafter, at locations not more than **900 millimeter 3 feet** from the ends of each runout, and not over **300 millimeter 1 foot** from each change in direction of piping.

Base load rating for all pipe-hanger supports on insulated weight of lines filled with water and forces imposed. Deflection per span is not exceed slope gradient of pipe. Ensure supports are in accordance with the following minimum rod size and maximum allowable hanger spacing for specified pipe. For concentrated loads such as valves, reduce the allowable span proportionately:

<u>PIPE SIZE (DN)</u> <u>MILLIMETER</u>	<u>ROD SIZE</u> <u>MILLIMETER</u>	<u>STEEL PIPE</u> <u>MILLIMETER</u>	<u>COPPER PIPE</u> <u>MILLIMETER</u>
25 and smaller	10	2500	1850
32 to 40	10	3050	2500
50	10	3050	3050
65 to 90	13	3700	3700
100 to 125	16	5000	4300
150	20	5000	5000
200 to 300	22	6100	6100
356 to 457	25	6100	6100
508 and over	32	6100	6100
<u>PIPE SIZE</u> <u>INCHES</u>	<u>ROD SIZE</u> <u>INCHES</u>	<u>STEEL PIPE</u> <u>FEET</u>	<u>COPPER PIPE</u> <u>FEET</u>
1 and smaller	3/8	8	6
1-1/4 to 1-1/2	3/8	10	8
2	3/8	10	8
2-1/2 to 3-1/2	1/2	12	12
4 to 5	5/8	16	14
6	3/4	16	16
8 to 12	7/8	20	20
14 to 18	1	20	20

PIPE SIZE
INCHES
20 and over

ROD SIZE
INCHES
1-1/4

STEEL PIPE
FEET
20

COPPER PIPE
FEET
20

Install vibration isolation supports where needed. Refer to Section 23 05 48.19 SEISMIC BRACING FOR HVAC where A/C equipment and piping is installed.

Support vertical risers independently of connected horizontal piping, whenever practicable, with fixed or spring supports at the base and at intervals to accommodate system range of thermal conditions. Ensure risers have guides for lateral stability. For risers subject to expansion, install only one rigid support at a point approximately one-third down from the top. Place clamps under fittings unless otherwise specified. Support carbon-steel pipe at each floor and at not more than 4572 millimeter 15-foot intervals for pipe DN50 2 inches and smaller and at not more than 6096 millimeter 20-foot intervals for pipe DN65 2-1/2 inches and larger.

3.4 PENETRATIONS

Install effective sound stopping and adequate operating clearance to prevent structure contact where piping penetrates walls, floors, or ceilings into occupied spaces adjacent to equipment rooms; where similar penetrations occur between occupied spaces; and where penetrations occur from pipe chases into occupied spaces. Occupied spaces include space above ceilings where no special acoustic treatment of ceiling is provided. Finish penetrations to be compatible with surface being penetrated.

- [Accomplish sound stopping and vapor-barrier sealing of pipe shafts and large floor and wall openings by packing to high density with properly supported fibrous-glass insulation or, where ambient or surface temperatures do not exceed 49 degrees C 120 degrees F, by foaming-in-place with self-extinguishing, 0.9 kilogram 2-pound density polyurethane foam to a depth not less than 152 millimeter 6 inches. Finish foam with a rasp. Ensure vapor barrier is not less than 3 millimeter 1/8-inch thick vinyl coating applied to visible and accessible surfaces. Where high temperatures and fire stopping are a consideration, use only mineral wool with openings covered by 1.6 millimeter 16-gage sheet metal.

]3.5 SLEEVES

Install sleeves where piping passes through roofs, masonry, concrete walls and floors.

Continuously [weld][brazed] sleeves passing through steel decks to the deck.

Ensure sleeves that extend through floors, roofs, load bearing walls, and fire barriers are continuous and fabricated from Schedule 40 steel pipe, with welded anchor lugs. Form all other sleeves by molded linear polyethylene liners or similar materials that are removable. Ensure diameter of sleeves is large enough to accommodate pipe, insulation, and jacketing without touching the sleeve and provides a minimum 10 millimeter 3/8-inch clearance. Install a sleeve size to accommodate mechanical and thermal motion of pipe precluding transmission of vibration to walls and the generation of noise.

Pack the space between a pipe, bare or insulated, and the inside of a pipe

sleeve or a construction surface penetration solid with a mineral fiber conforming to **ASTM C553** Type V (flexible blanket), **to 538 degrees C to 1,000 degrees F**. Install this packing wherever the piping passes through firewalls, equipment room walls, floors, and ceilings connected to occupied spaces, and other locations where sleeves or construction-surface penetrations occur between occupied spaces. Where sleeves or construction surface penetrations occur between conditioned and unconditioned spaces, fill the space between a pipe, bare or insulated, and the inside of a pipe sleeve or construction surface penetration with an elastomer caulk to a depth of **13 millimeter 1/2 inch**. Ensure all caulked surfaces are oil- and grease-free.

Ensure through-penetration fire stop materials and methods are in accordance with **ASTM E814** and **UL 1479**.

Caulk exterior wall sleeves watertight with lead and oakum or mechanically expandable chloroprene inserts with mastic-sealed metal components.

NOTE: Review roof flooding provisions before revising the following paragraph.

[Ensure sleeve height above roof surface is a minimum of **305 12** and a maximum of **457 millimeter 18-inches**.

]3.6 ESCUTCHEONS

Install escutcheons at all penetrations of piping into finished areas. Where finished areas are separated by partitions through which piping passes, install escutcheons on both sides of the partition. Where suspended ceilings are installed, install plates at the underside only of such ceilings. For insulated pipes, select plates large enough to fit around the insulation. Use chrome-plated escutcheons in all occupied spaces and of size sufficient to effectively conceal openings in building construction. Firmly attach escutcheons with setscrews.

3.7 FLASHINGS

NOTE: Review roof flooding provisions.

[Install flashings at penetrations of building boundaries by mechanical systems and related work.

]3.8 UNDERGROUND PIPING INSTALLATION

Prior to being lowered into a trench, clean all piping, visually inspected for apparent defects, and tapped with a hammer to audibly detect hidden defects.

Further inspect suspect cast-ferrous piping by painting with kerosene on external surfaces to reveal cracks.

Distinctly mark defective materials found using a road-traffic quality yellow paint; promptly remove defective material from the site.

After conduit has been inspected, and not less than 48 hours prior to

being lowered into a trench, coat all external surfaces of cast ferrous conduit with a compatible bituminous coating for protection against brackish ground water. Apply a single coat, in accordance with the manufacturer's instructions, to result in a dry-film thickness of not less than 0.30 millimeter 12 mils.

Ensure excavations are dry and clear of extraneous materials when pipe is being laid.

Use wheel cutters for cutting of piping or other machines designed specifically for that purpose. Electric-arc and oxyacetylene cutting is not permitted.

Begin laying of pipe at the low point of a system. When in final acceptance position, ensure it is true to the grades and alignment indicated, with unbroken continuity of invert. Blocking and wedging is not permitted.

[Point bell or grooved ends of piping upstream.

] Make changes in direction with long sweep fittings.

Install necessary socket clamping, piers, bases, anchors, and thrust blocking. Protect rods, clamps, and bolting with a coating of bitumen.

Support underground piping below supported or suspended slabs from the slab with a minimum of two supports per length of pipe. Protect supports with a coating of bitumen.

On excavations that occur near and below building footings, install backfilling material consisting of 13800 kilopascal 2,000-psi cured compressive-strength concrete poured or pressure-grouted up to the level of the footing.

Properly support vertical downspouts; soil, waste, and vent stacks; water risers; and similar work on approved piers at the base and provided with approved structural supports attached to building construction.

[Provide cleanout, flushing, and observation risers.

]3.9 HEAT TRACE CABLE INSTALLATION

Field apply heater tape and cut to fit as necessary, linearly along the length of pipe after piping has been pressure tested and approved by the Contracting Officer. Secure the heater to piping with [cable ties] [fiberglass tape]. Label thermal insulation on the outside, "Electrical Heat Trace."

Install power connection, end seals, splice kits and tee kit components in accordance with IEEE 515 to provide a complete workable system. Terminate connection to the thermostat and ends of the heat tape in a junction box. Ensure cable and conduit connections are raintight.

3.10 DISINFECTION

[Disinfect water piping, including all valves, fittings, and other devices, with a solution of chlorine and water. Ensure the solution contains not less than 50 parts per million (ppm) of available chlorine. Hold solution for a period of not less than 8 hours, after which the solution contains

not less than 10 ppm of available chlorine or redisinfect the piping. After successful sterilization, thoroughly flush the piping before placing into service. Flushing is complete when the flush water contains less than 0.5 ppm of available chlorine. Water for disinfection will be furnished by the Government. Approve disposal of contaminated flush water in accordance with written instructions received from the Environmental authority having jurisdiction through the Contracting Officer and all local, State and Federal Regulations.

] [Flush piping with potable water until visible grease, dirt and other contaminants are removed (visual inspection).

] 3.11 HEAT TRACE CABLE TESTS

Test heat trace cable system in accordance with IEEE 515 after installation and before and after installation of the thermal insulation. Test heater cable using a [1000] [_____] vdc megger. Minimum insulation resistance is [20 to 1000] [_____] megohms regardless of cable length.

3.12 OPERATION AND MAINTENANCE

Provide Operation and Maintenance Manuals consistent with manufacturer's standard brochures, schematics, printed instructions, general operating procedures and safety precautions. Submit test data that is clear and readily legible.

3.13 PAINTING OF NEW EQUIPMENT

NOTE: For SOUTHNAVFACENGCOM projects, delete all painting requirements and specify as follows: "PART 3 EXECUTION, Not used."

Factory or shop apply new equipment painting, as specified herein, and provided under each individual section.

3.13.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied withstands 125 hours in a salt-spray fog test, except that equipment located outdoors withstand 500 hours in a salt-spray fog test. Conduct salt-spray fog test in accordance with ASTM B117, and for that test the acceptance criteria is as follows: immediately after completion of the test, the inspected paint shows no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shows no signs of rust creepage beyond 3 mm 0.125 inch on either side of the scratch mark.

Ensure the film thickness of the factory painting system applied on the equipment is not less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C 120 degrees F, design the factory painting system for the temperature service.

3.13.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the

surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal, surfaces subject to temperatures in excess of 50 degrees C 120 degrees F.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Selected color of finish coat is aluminum or light gray.

- a. Temperatures Less Than 50 Degrees C 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C 120 degrees F receives one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm 0.3 mil, one coat of primer applied to a minimum dry film thickness of 0.0255 mm one mil; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm one mil per coat.
- b. Temperatures Between 50 and 205 Degrees C 120 and 400 Degrees F: Metal surfaces subject to temperatures between 50 and 205 degrees C 120 and 400 degrees F Receives two coats of 205 degrees C 400 degrees F heat-resisting enamel applied to a total minimum thickness of 0.05 mm 2 mils.
- c. Temperatures Greater Than 205 Degrees C 400 Degrees F: Metal surfaces subject to temperatures greater than 205 degrees C 400 degrees F receives two coats of 315 degrees C 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm 2 mils.

-- End of Section --